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B.Sc. (Hons.) SEMESTER V EXAMINATION 2016-17**PHYSICS****BPT - 505 : Electromagnetic Theory****Time : Three hours****Max. Marks : 70****(WRITE YOUR ROLL NO. AT THE TOP IMMEDIATELY ON THE RECEIPT OF THIS QUESTION PAPER)***The figures in the right-hand margin indicate marks***Answer all questions***Mathematical symbols used in the text carry their usual meaning***1. Answer all questions****4 × 5**

- (a) Show that the vector field $\vec{v} = \hat{a}_x yz + \hat{a}_y zx + \hat{a}_z xy$ can be written as a gradient of a scalar field $f(x, y, z)$.
- (b) If an infinitely long straight wire carries a uniform line charge ρ_L , what would be the electric field at a distance x from this wire.
- (c) If a potential due to a continuous charge distribution is written as $V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{R} dv$, obtain $\nabla^2 V(\vec{r})$.
- (d) A polarized wave is incident from air to polystyrene with $\mu = \mu_0, \epsilon = 2.6\epsilon_0$ at Brewster angle. Determine the transmission angle.
- (e) Determine the state of the polarization for the wave whose electric field is given by $\vec{E}(z, t) = E_0 [\hat{a}_x \cos(kz - \omega t) - \hat{a}_y \sin(kz - \omega t)]$.
- (f) Calculate the intrinsic impedance of the free space.

— (2) —

2. (a) What is a Linear medium? How do the Maxwell's equations modify in such a linear dielectric? **6**
- (b) Does the Poynting vector correspond to dual character in the energy momentum conservation in electrodynamics? If so, explain. **6**
3. (a) In a lossless medium for which $\eta = 60\pi$, $\mu_r = 1$, and $\vec{H} = -0.1 \cos(\omega t - z)\hat{a}_x + 0.5 \sin(\omega t - z)\hat{a}_y$ A/m, find the values of ϵ_r and ω ? **6**
- (b) What is a phasor? The electric and magnetic fields in free space are given by $\vec{E} = \frac{50}{\rho} \cos(10^8 t + \beta z)\hat{a}_y$ V/m and $\vec{H} = \frac{H_0}{\rho} \cos(10^8 t + \beta z)\hat{a}_x$ A/m. Find the phasor form of electric and magnetic fields. **6**
4. (a) How do the laws of electrostatics enforce boundary conditions on electric field and the corresponding potentials? **6**
- (b) Why does any net charge resides on the surface of a conductor? Explain. **6**

OR

- (a) Using the general solution of Laplace equation in the spherical polar co-ordinate with azimuthal symmetry, determine the potential outside the surface of a hollow empty sphere of radius R if the surface potential is $V_0(\theta) = k \sin^2(\theta/2)$. k is a parameter. **6**
- (b) For a uniform $\vec{B}$, show that $\vec{A} = -\frac{1}{2}(\vec{r} \times \vec{B})$ such that $\vec{\nabla} \cdot \vec{A} = 0$ and $\vec{\nabla} \times \vec{A} = \vec{B}$. **6**
5. (a) A standard air filled rectangular waveguide with dimensions $a = 8.636$ cm, $b = 4.318$ cm is fed by a 4-GHz carrier wave from a co-axial cable. Determine if a TM_{11} mode will be propagated. If so, calculate the phase and group velocities. **6**
- (b) Show that a rectangular waveguide does not support TM_{10} and TM_{01} modes. **8**

OR

- (a) Derive the general expressions of the attenuation (α) and phase constants (β) for the conducting media. **8**
- (b) Prove that $1 + \Gamma_{\parallel} = \tau_{\parallel} \left(\frac{\cos \theta_t}{\cos \theta_i} \right)$, where all symbols have their usual meaning. **6**

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- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app>